2.4.2 Tradeoffs in identity naming

When we pick a naming system, there are several tradeoffs we must manage depending on our goals, and different naming systems prioritize different aspects:

- **Human Meaningfulness**—Some use cases require names that are easy for people to recognize and remember. Examples include email addresses and usernames.
- **Uniqueness and Non-Reusability**—Some systems require non-reusable identifiers to prevent conflicts and identity takeover. One example is employee IDs, which should not be reassigned.
- **Security Against Impersonation and Forgery**—Some use cases are sensitive to spoofing and collisions and have differing security needs. Domain names are a good example where having proper security against hijacking is vital to their use.
- **Decentralized**—Some use cases need names that don't have to be managed by a central authority. For example, UUIDs (Universal Unique Identifiers) are created in such a way that different systems don't have to coordinate to be assured that the identifier they're using is globally unique.

Different systems must balance these competing concerns to get the best naming system for their expected use.

2.4.3 Namespaces and their role in identity

As I said earlier, a namespace is the context for a name. But it's more than just that: A namespace is a system for structuring names and ensuring their uniqueness within that context. We can classify namespaces based on how they prevent name collisions:

- **Centralized**—A centralized namespace is maintained by a particular organization, which ensures identifier uniqueness. The primary advantage is the ease of implementation. But names in centralized namespaces aren't easily portable. Usernames and employee ID numbers are both examples of centralized namespaces.